

GOUTHAM REDDY KONDREDDY

Performance Engineer / GPU-Kernels
Optimization Engineer — AI Research

CONTACT

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LINKS

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SKILLS

Languages

C++, CUDA, C, Python

GPU / Parallel

CUDA kernel development, tensor-core MMA, shared-memory tiling, warp-shuffle reductions, cp.async pipelines; CPU SIMD (AVX2), OpenMP, multithreading

ML / Systems

Deep-learning framework & autograd internals (PyTorch / TensorFlow), computational-graph capture, BLAS / GEMM, mixed precision, Nsight Systems/Compute profiling, DL-compiler concepts

EDUCATION

B.Tech, Computer Science & Engineering

Sri Venkateswara University · 2021–2025 · CGPA 8.02 / 10

INTERESTS

Foundation-model & LLM training infrastructure · GPU performance engineering · vision / audio & multimodal models

SUMMARY

CUDA kernel engineer building a deep-learning framework from first principles. I design and optimize the low-level tensor-ops and BLAS layer — reductions, datatypes, memory management, and GEMM — for large-language-model training, with a focus on matching and beating production-framework throughput on NVIDIA GPUs.

EXPERIENCE

BluBridge Technologies — Chennai

AI Research Intern · Tensor & Ops team (GPU kernel optimization) · 2025–Present

- Built the framework's **reduction module** — 22 unified GPU/CPU reduction ops (sum/prod/min/max/mean, NaN-aware, argmin/argmax, variance/std, fused var-mean) behind a single templated kernel via a 2-level dispatch design, replacing ~800 lines of duplicated kernels (warp-shuffle + shared-memory reduction on GPU; OpenMP + cascade-sum on CPU).
- Implemented the **datatype system**: FP16 / BF16 / FP8 (E4M3, E5M2), complex (32/64/128) and integer types, with bit-level FP32 ↔ BF16/FP16 conversion and **dtype-promotion** logic integrated across all tensor operations.
- Implemented a **TensorFlow-style BFC memory allocator** and restructured the library's allocation / deallocation / memcpy / memset layer.
- Led a **kernel-optimization sprint** on a 124M-parameter GPT-2 training workload (Karpathy nanoGPT config): cuBLASLt matmul port + fused-bias epilogue, stride-aware attention, Packed SDPA, vectorized LayerNorm/Gelu, and memory-traffic elimination — **closing a ~13.5 s/run gap to PyTorch and surpassing it on some runs**.
- Currently developing **Level-2/3 BLAS GEMM kernels for NVIDIA Blackwell (B300, sm_100)** to efficiently train sub-billion → multi-billion-parameter LLMs.

CEERAS IT Solutions

Co-Founder (student venture) · 2024

- Co-founded a small student technology venture; led operations and delivery.

LEADERSHIP

Placement Coordinator · Chief Operating Officer, Ignite Club · Class Representative (2 semesters).